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Factors affecting duration and effort estimation errors in software development projects

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Abstract

The purpose of this research was to fill a gap in the literature pertaining to the influence of project uncertainty and managerial factors on duration and effort estimation errors. Four dimensions were considered: project uncertainty, use of estimation development processes, use of estimation management processes, and the estimator's experience. Correlation analysis and linear regression models were used to test the model and the hypotheses on the relations between the four dimensions and estimation errors, using a sample of 43 internal software development projects executed during the year 2002 in the IT division of a large government organization in Israel. Our findings indicate that, in general, a high level of uncertainty is associated with higher effort estimation errors while increased use of estimation development processes and estimation management processes, as well as greater estimator experience, are correlated with lower duration estimation errors. From a practical perspective, the specific findings of this study can be used as guidelines for better duration and effort estimation. Accounting for project uncertainty while managing expectations regarding estimate accuracy; investing more in detailed planning and selecting estimators based on the number of projects they have managed rather than their cumulative experience in project management, may reduce estimation errors.
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Keyword: Duration and effort estimation errors; Software development projects

1. Introduction

In today's fast changing world, where success in managing projects is a critical factor for the success of the entire organization, and many projects are terminated when it becomes apparent that they will significantly exceed their planned time and budget goals [8], accurate estimation of project duration and required resources becomes an issue of prime importance. On the one hand, overestimating time and effort (or budget), due to a presumed lack of resources or because the projected completion is too late, can convince management not to approve projects that may otherwise contribute to the organization. On the other hand, underestimation may result in approval of projects that will fail to deliver the expected product within the time and budget available. In spite of the critical role of accuracy, examples of incorrect estimation abound, especially in IT projects [23], resulting in enormous waste of time and money. Most studies dealing with estimation errors focus on a single group of factors affecting the accuracy of estimation. Very few are based on actual project data, which is hard to come by. The purpose of the study reported here was to identify the main factors affecting time and effort estimations and to propose guidelines for improving estimation results in IT projects.

2. Literature review

Task duration and effort estimates are determined during the planning stage of the project and provide the basis for subsequent planning, control, and decision making. We use the term *estimation error* to denote the

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difference between the estimates obtained during planning and the actual time or effort required for completing the task. There are two types of estimation errors: overestimation – when the actual results exceed the planned values, and underestimation – when the actual results fall below the planned values. Both types are harmful and can damage the organization's reputation and management support.

2.1. The effect of uncertainty on estimation accuracy

Estimates are in effect, guesses regarding future performance based on available knowledge. As such, their accuracy is affected by the extent of uncertainty regarding the task to be estimated. Uncertainty is associated, inter alia, with definitions of requirements, choice of technological solutions, innovativeness of need, and customer characteristics.

Analyzing the relation of time, cost and effort estimations, with uncertainty in large software projects, Hihn and Habib-aghi [15] found that when the variance in time and budget was large, the estimations were below the actual values. The cost estimations in the studies they examined usually used some kind of a known model with modifications for innovativeness, complexity and staffing of the project. Project size was also found to affect time estimations; in small projects, communication is tighter and problems are easily dealt with and therefore, time estimation errors are smaller [3]. Inaccuracy in estimation and hence reduced project performance is often a result of uncertainty and not a result of execution problems [10,26].

2.2. Estimation development

One of the main factors affecting estimation accuracy is the level of detail in preparing the project plan. Although intuitively the bottom-up approach seems to provide better data for estimation, empirical studies have shown that the selection of the proper method depends on the task characteristics, and the level of the estimators' experience [13,16]. Underestimation is common to complex tasks that have not been sufficiently broken down [14]. When estimation errors are independent, dividing the project into smaller work packages provides better data for estimation and reduces the statistical errors [28].

Since it is almost impossible to accurately estimate the project duration at the very start, accuracy is improved by an iterative process throughout the life cycle of the project [17].

Managerial control was found in several studies to have a high and positive effect on estimation accuracy, and the inter-relations between the development team and the customer has emerged as the most important among the managerial factors [29]. A comprehensive causal model presenting the relationship between estimation errors and three categories of managerial factors was proposed by Lederer and Prasad [24]:

1. *Estimation goals*. Goals that support project execution, such as project staffing, monitoring and control, scheduling and project selection, and goals that support the estimation process.
2. *Estimation management processes*. Preparation and updating processes (definition, design, feasibility study, and change control), the level of responsibility assumed by the IS department, and responsibility for and commitment to the customer, as demonstrated in design reviews and throughout the project.
3. *Estimation techniques*. Algorithmic techniques (statistical formulas, differential equations, and organizational standards) and informal techniques such as intuition, comparison with earlier projects and guessing.

The managerial processes determine the choice of the estimation techniques; for example, estimation assessment reviews encourage the use of algorithmic techniques (which are assumed to be more accurate than informal techniques). The processes that were found to contribute most to estimation accuracy were those that aimed at enhancing the sense of responsibility and commitment among the IS personnel.

2.3. Estimation management

Estimation accuracy is not only a technical problem that can be handled by using computational models [12]. It is also a managerial problem that should be treated by creating motivation and commitment to success. As a managerial problem, estimation accuracy is influenced by managerial decisions related to the estimation goals and the way estimation is carried out and updated. Estimation is performed for a variety of reasons: project selection, staffing, scheduling, monitoring and control, team performance assessment, and marketing. Management and client involvement, as well as political/organizational factors such as the inability to accurately determine the programmers' skill level, lack of estimation methods, pressures to lower the estimation, lack of past data, and frequent changes by the users may all have a substantial influence on the estimation accuracy [22].

2.4. Estimator's experience and estimation accuracy

Estimator's experience is considered one of the most important factors affecting estimation accuracy. Most researchers agree that the estimator's specific qualifications are more important when expert opinion or analogy are used than in formal techniques, which are considered more objective. Nevertheless, even when such techniques are used, e.g. COnstructive COst MOdel (COCOMO), an expert opinion is required for selection of the proper input variables. External estimators provide consistently lower estimations than the developers themselves [22], and in many cases there is a gap between the estimators' perception of the required skill and the actual skill used in the project [5].

2.5. Estimation techniques

The proper selection and application of estimation techniques is a main factor affecting estimation accuracy. Estimation techniques described in the project management professional literature can be divided into the following categories:

Estimation by analogy. The effort and time to complete a certain task are estimated by comparison to similar tasks done before. The main problem of this low cost, simple but reasonably accurate technique is the difficulty in finding suitable analogies [11]; previous cases are usually not similar enough to the case under consideration, and an expert is required to bridge the gap [31]. Furthermore, company-specific models seem not to yield better results than the external models [30].

Parametric models. Models that use historical data to identify the main factors affecting time and effort estimations. The main parameters usually used in such models are system size, complexity, skills and experience of the project personnel, hardware limitations, software development tools, stability of user requirements, and reuse of SW routines. Different models use the same factors in different ways: COCOMO defines project size by the number of code lines [2], while the Function Point model uses the number of entities and their complexity [1]. However, it was found lately that COCOMO's estimates can be significantly improved using a feature subset selection method developed by the data mining community [6].

Expert estimation is the most frequently applied estimation strategy for software projects [20], due to its simplicity and low cost. Studies show that expert estimation can be fairly accurate but it is hard to identify the factors contributing to the estimation and to isolate the experience component from the expert subjectivity [19]. Expert estimation can be influenced by the estimator's experience and his involvement in the development effort [1]. Estimators' optimism may lead to underestimation when estimators are in charge of the task [7]. A possible explanation is the "planning illusion" [4], where estimation is based on future understanding of the way the task should be performed without learning from similar previous tasks. Estimators tend to provide underestimations for tasks perceived as easy, and overestimations for those perceived as difficult [27]. There are situations where expert estimates can be expected to be more accurate than formal estimation models [20].

Artificial Intelligence techniques. Like case-base reasoning, which is based on analogy with past projects, the system computes the differences between the current project and past projects to select similar projects for use as the base for analogy [25].

A study of 364 organizations [12] found that less than 150 used models for effort estimation, but those organizations that did not use models performed no worse than others. Another study conducted in the Netherlands [18] found that the use of parametric models was quite rare (14%), expert estimations were used in about 25% of the cases, and analogy was the most widely used (61%). The difficulties in using models are mainly the level of effort required to collect historic data and keeping the databases updated. An analysis of four parametric models has shown errors in the range of 85–772% [21]. Comparison of four estimation techniques found that expert estimation was as accurate as the other three: case-based reasoning, COCOMO, and function point [9].

The objective of this study is to provide specific guidance on improving estimation performance and reducing estimation errors. To this end, several hypotheses based on the research literature were developed and tested using data from a sample of IT projects carried out by a large governmental organization. The paper is organized as follows. The next section presents the variables and hypotheses. The following section describes the sample characteristics. Then, we present the statistical analysis that was carried out in order to test the hypotheses, following with a discussion and conclusions on how to reduce estimation errors. We conclude with a discussion of the limitations of this study and possible avenues for further research.

3. Research hypotheses

Based on the literature review we identified four main dimensions that affect estimation accuracy: project uncertainty, estimation development processes, estimation management processes, and estimator experience.

The amount of uncertainty perceived by those who had to estimate the duration of the project tasks and the effort required to carry them out was measured by several variables affecting the difficulty of assessing time and effort required to perform the project tasks, such as its overall size (in terms of forecasted duration or person-months), its managerial complexity (as indicated by team size, number of users, and implementation complexity), innovativeness of need, the experience of the team, and the anticipated availability of resources.

By estimation development we mean the various processes that contribute, either directly or indirectly, to the generation of the actual duration and effort estimates. These include defining the purpose and intended use of the estimates, developing a detailed Work Breakdown Structure (WBS), applying estimation techniques such as those reviewed in the previous section, and involving management, team members and other stakeholders. Estimation management included controlling actual performance against estimates and updating as appropriate, carrying out systematic risk assessments to validate the estimates, and implementing managerial policies that promote the commitment of team members to the estimates that constitute their project plan. Estimator experience was measured in terms of years of experience and number of projects with similar technologies and systems the estimator was involved with.

In summary, our assumption is that the relative magnitude of duration and effort estimation errors is affected by the extent of uncertainty in the project, the amount of effort invested in developing and managing the estimates, and by the experience of the estimators.

Based on this conceptual framework, the following four broad hypotheses were derived:

H1. Projects with higher level of uncertainty exhibit larger duration and effort estimation errors.
H2. Projects with higher use of estimation development processes exhibit smaller duration and effort estimation errors.
H3. Projects with higher use of estimation management processes exhibit smaller duration and effort estimation errors.
H4. Projects where the estimates were made by more experienced estimators will have smaller duration and effort estimation errors.

The four hypotheses pertain to errors in estimating both the duration of the project tasks and the amount of effort they require. Hypotheses testing were carried out on the factors that constitute each of the four main constructs, as described in the next section.

4. Method

4.1. The sample

Data for this study were collected from the IT division of a large government organization in Israel that employs several hundred people and has an annual budget in the order of tens of millions of US dollars. The division is engaged in a variety of software development projects, including both application and infrastructure systems, fully or partially outsourced, ranging from implementation of commercial off-the-shelf software, through integration of software packages, up to in-house development of new systems. The sample for the analysis consisted of the internal software development projects executed during the year 2002 and completed up to the end of June 2003 – altogether 47 projects. For each project, data were collected from two sources: measurement data from the division Enterprise Project Management Information System (EPMIS), and a questionnaire that was developed specifically for this study and was administered to the individuals who managed the 47 projects.

4.2. Questionnaire data

The questionnaire developed for the research was pilot tested for content validity on a sample of five project managers in the organization. The final version of the questionnaire was used to collect data from the 24 individuals who managed the projects in the sample (some individuals managed more than one project). Each respondent filled out the questionnaire at an individual meeting with one of the researchers, during which there was ample opportunity to clarify the questions and to verify the answers. Respondents were assured of the confidentiality of their answers. Each meeting lasted between 60 and 90 min, and yielded one completed questionnaire for each of the projects in the sample.

4.3. Structure of the questionnaire

The questionnaire consisted of two types of questions. The first type included factual questions relating to the project (team size, duration, budgeted effort, number of users, type of technology, control frequency, etc.) and to the respondent position in the organization (years of experience, previous training, etc.). The second type, which was the majority, consisted of statements regarding the project. The respondents were asked to indicate the extent to which they agreed with each statement on a 1–5 Likert scale. These statements are listed in Appendix A. The questionnaire was structured in four main sections corresponding to the four main dimensions that, according to the literature survey, influence estimation accuracy: the project uncertainty, the estimation development, the estimation management processes, and the estimator's experience.

Data from the questionnaires were merged with data from the EPMIS by project. An initial validity check of the data revealed four projects where the estimation errors exceeded the sample mean by more than three standard deviations. These large estimation errors were found to be a result of incorrect data entry into the EPMIS. Consequently, these four projects were excluded, yielding a sample of 43 valid projects.

4.4. Descriptive statistics

Descriptive statistics for selected relevant variables based on the final sample of 43 projects are presented in Table 1.

From the table we note that most of the projects in the sample were relatively small in terms of team size, duration and total effort. More than 90% of the projects were planned to last one year or less. Although some projects were broken down into four WBS levels, the relatively low average value of 2.14, combined with average task duration of 75 days indicates that in general planning was not very detailed. This might be due to the fact that in the organization formal progress reporting to management was required on a quarterly basis, and thus most project managers had no external incentive to plan to a much finer level.

In 77% of the projects, monitoring and control was done weekly or bi-weekly. This frequency of control is a result of the small size of most of the projects, which typically last less than a year. Customer control was less frequent, and usually done (in 93% of the cases) by a steering committee. Customer control was tied to major milestones, such as Preliminary Design Reviews (PDR), Critical Design Reviews

Table 1
Descriptive statistics for the projects in the sample

Variable	Mean	Standard deviation	Median	Max	Min	N
Team size (persons)	3.28	2.15	3.00	11.00	1.00	43
Estimated total duration (months)	9.17	3.59	6.00	17.00	3.00	43
Estimated total effort (person-hours)	1734.80	1400.82	713.00	6770.00	310.00	43
Number of end users	908.88	1417.28	200.00	4500.00	5.00	43
Actual project duration (months)	11.53	4.23	12.00	24.00	3.00	43
Actual total effort (person-hours)	2071.93	1825.71	1375.00	9777.00	360.00	43
Average number of WBS levels	2.14	.71	1.83	4.00	1.00	43
Average task duration (days)	75.19	53.80	59.00	225.00	6.00	42
Average task effort (person-hours)	205.18	187.48	153.00	985.00	5.00	42

(CDR), end of prototype development, etc. The project plan was usually updated on a quarterly basis (77%) as required by the organization's procedures.

Estimator experience was measured in several ways: four Likert scale items, four items pertaining to years of experience, two binary items regarding previous training, a question on the number of projects previously managed and a question on the position of the respondent in the organization. The number of projects that the estimator had managed ranged from 1 to 20, with an average of 8.42 and a standard deviation of 5.32. Table 2 summarizes the estimators' level of experience.

It seems that in general the estimators in the sample were not novices and had acquired a substantial amount of experience either as project managers, team leaders or team members in their technological environment.

4.5. Estimation errors

In 32 of the 43 projects the actual duration exceeded the estimated duration; in four projects the estimated duration exceeded the actual duration, and in the remaining seven projects the project was completed as planned. With respect to effort estimation, 32 cases resulted in underestimation (not necessarily the same 32 cases in which the duration was underestimated), and 11 cases resulted in effort overestimation. Since in this study we were concerned with the factors that affect the size of the estimation errors rather than the direction of the error, we did not distinguish between overestimations and underestimations. For both duration and effort we defined as dependent variable the magnitude of the estimation error relative to the actual value (DRE and ERE). Table 3 presents the descriptive statistics for the two dependent variables in the sample of 43 projects.

On the average, the duration estimation error was about two and a half months, or about 22% of the entire project duration. The average effort error was about three person months (based on 150 h per month), or about 24% of the entire project estimated effort. In both cases, the errors are substantial but not extremely large. In fact, within the software development industry estimation errors of 20–25% are not unusual and may even be considered acceptable. The median values were in both cases slightly less than the average, indicating slight skew to the right.

Table 2
Estimators level of experience

Years of experience	<2	3–5	6–7	8–9	>10
In the organization	2	11	8	14	8
In software development	0	6	4	5	28
Managing software development projects	4	10	9	11	9
Estimating software development projects	4	14	9	7	9

Table 3
Descriptive statistics for the dependent variables

Dependent variable	Average	Standard deviation	Median	Min	Max
Duration relative error – DRE (%)	22	17	20	0	75
Effort relative error – ERE (%)	24	17	22	2	76

5. Statistical analysis

5.1. Factor analysis

The first step in the analysis consisted of factor analysis of the Likert scale questionnaire items. Some of the items were recoded in reverse (i.e., response value 5 was changed to 1, 4 to 2, and so on) in order to ensure semantic consistency within each construct. The analysis was done in an iterative manner separately for each group of questionnaire items associated with the four constructs in the model. In every iteration, items that failed to reach the threshold loading of 0.5 and those that had high loadings on multiple factors were excluded. In addition, items that were assigned to factors to which they clearly did not belong in terms of their meaning were also excluded. The process stopped when no more items were excluded. Then, within each identified factor a reliability analysis was performed based on the Cronbach α coefficient. A minimum threshold value of 0.6 was set as recommended by Van de Ven and Ferry [29] and in line with similar studies [24]. The final list of factors appears in Table 4, along with their respective values of the Cronbach α coefficients and the loadings of the questionnaire items. After the data reduction and reliability analysis, the value for each factor was calculated as a simple average of the questionnaire items it included. This average served as the factor score in the correlation analysis described in the following section.

Table 4
Final list of factors

Dimension	Factor definition (Cronbach α)	Questionnaire items	Loadings
Project uncertainty	Unclear project definition (0.8210)	1.2.1	0.821
		1.2.2	0.891
	Low project importance (0.8884)	1.2.3	0.688
		1.2.5	0.793
	Technological uncertainty (0.7005)	1.3.1	0.838
		1.3.2	0.702
		1.3.3	0.727
	Implementation complexity (0.8604)	1.3.5	0.908
		1.3.6	0.861
		1.3.7	0.787
	Low team experience (0.8098)	1.4.1	0.623
		1.4.2	0.816
		1.4.3	0.877
		1.4.4	0.663
	Unavailability of resources	1.4.9	0.864
	Instability (0.6216)	1.2.4	0.738
		1.5.1	0.790
		1.5.4	0.589
	Lack of client preparedness (0.7477)	1.5.2	0.632
		1.5.3	0.816
		1.5.5	0.633
		1.5.6	0.690
Estimation development	Estimation goals definition (0.7280)	2.1.4	0.578
		2.1.5	0.861
		2.1.6	0.800
	Team focused processes (0.6231)	2.1.3	0.595
		2.1.16	0.774
		2.1.18	0.727
	Participation of other groups	2.1.17	0.858
Estimation management	Customer control (0.8421)	2.2.2	0.885
		2.3.7	0.764
	IT unit control (0.7125)	2.2.4	0.819
		2.3.6	0.855
	Reporting frequency (0.6717)	2.3.5	0.912
		2.3.9	0.678
	Project team performance assessment (0.8572)	2.3.11	0.917
		2.3.12	0.899
	Systematic risk assessment (0.6389)	2.4.2	0.776
		2.4.3	0.834
Estimator experience	Estimator experience in the application area (0.8190)	3.1.9	0.838
		3.1.10	0.839
		3.1.11	0.823
		3.1.12	0.733

5.2. Correlation analysis

The Pearson coefficients of correlation were calculated for independent variables measured on numerical scales, while the Spearman correlation coefficients were calculated for variables measured on an ordinal scale. The results for each of the four dimensions of our framework are presented next.

Table 5 shows the Pearson coefficients of correlation between the project uncertainty factors of Table 4 and the two measures of estimation errors. The last row shows the Spearman coefficient of correlation with the ordinal scale variable, innovativeness of need, consisting of five classes, from minor extensions to existing system to pilot projects. The estimated total duration and low team experience are positively and significantly correlated with DRE, while implementation complexity is negatively correlated with DRE. Innovativeness of need is also correlated relatively high with DRE (.283), and the number of end users with ERE (.252). These correlations are not statistically significant probably due to the small sample size.

Table 5
Correlation coefficients for the project uncertainty factors ($N = 43$)

Independent variable	DRE	ERE
Team size(†)	−.202	−.013
Estimated total duration(†)	−.518**	.072
Estimated total effort(†)	−.133	−.153
Number of end users(†)	.075	.252
Unclear project definition	−.157	−.060
Low project importance	.205	.047
Technological uncertainty	.020	.116
Implementation complexity	−.290*	.067
Low team experience	.287*	.091
Unavailability of resources	.204	.031
Instability	−.057	−.050
Lack of client preparedness	.191	.204
Innovativeness of need	.283	.200

The variables indicated by (†) are factual items from the questionnaire; the other are aggregated variables obtained by factor analysis.
* Correlation is significant at the 0.05 level (one-tailed).
** Correlation is significant at the 0.01 level (one-tailed).

Table 6
Correlation coefficients for the estimation development factors ($N = 43$)

Independent variable	DRE	ERE
Number of WBS levels(†)	.199	−.118
Average task duration(†)	−.167	.445**
Average task effort(†)	−.011	.057
Estimation goals definition	−.337*	−.171
Team focused processes	−.181	.117
Participation of other groups	−.206	.150

The variables indicated by (†) are factual items from the questionnaire; the others are aggregated variables obtained by factor analysis.
* Correlation is significant at the 0.05 level (one-tailed).
** Correlation is significant at the 0.01 level (one-tailed).

Table 6 shows the Pearson coefficients of correlation between the estimation development factual variables[1] and estimation development factors and the relative estimation errors. The average task duration is significantly and positively correlated with ERE while the estimation goals definition factor is negatively correlated with DRE.

Table 7 shows the Pearson coefficients of correlation between the estimation management factual variables and estimation management factors and the relative estimation

[1] Factual variables are obtained from the plan of record, and as such they are factual observations, as opposed to subjective responses to questionnaire items.

Table 7
Correlation coefficients for the estimation management factors ($N = 43$)

Independent variables	DRE	ERE
Frequency of project manager control(†)	−.183	−.076
Frequency of steering committee control(†)	−.093	−.150
Frequency of IT unit control ($N = 40$)(†)	−.281*	−.184
Frequency of work plan update(†)	−.259*	.223
Customer control ($N = 40$)	−.066	−.405**
IT unit control	−.195	.071
Reporting frequency	−.369**	.032
Project team performance assessment	−.413**	.047
Systematic risk assessment	−.322*	−.023

The variables indicated by (†) are factual items from the questionnaire; the others are aggregated variables obtained by factor analysis.
* Correlation is significant at the 0.05 level (one-tailed).
** Correlation is significant at the 0.01 level (one-tailed).

errors. The reporting frequency, project team performance assessment and systematic risk assessment factors are significantly and negatively correlated with DRE and the customer control factor is negatively correlated with ERE. Frequency of control and frequency of work plan update are also negatively and significantly correlated with DRE (−.281 and −.259, respectively), but not so high as the other variables mentioned above.

Table 8 shows the correlations of the single factor extracted from the questionnaire items measured on the Likert scale and the ordinal variables used to measure experience with the two measures of estimation error. Only one significant correlation was found between the number of projects previously managed by the estimator and DRE. However, the estimator experience in the application area has also relatively high negative (−.254) correlation with DRE.

T-tests that were performed to find if there is a difference between average estimation errors provided by estimators with training in IT project management ($N = 20$) and those without such training ($N = 23$) show a significant difference (.045) only on effort estimation. When the estimators have some training in estimation techniques ($N = 11$), all their average errors are smaller than those of untrained estimators ($N = 32$). The difference for duration estimation errors is significant (.001).

Table 8
Correlation coefficients for the estimator experience factor ($N = 43$)

Independent variable	DRE	ERE
Number of projects previously managed(†)	−.285*	.173
Experience in the organization(†)	−.026	.034
Experience in software development(†)	−.127	.138
Experience managing software development projects(†)	.008	.089
Experience estimating software development projects(†)	.048	.097
Estimator experience in the application area	−.254	−.138

The variables indicated by (†) are factual items from the questionnaire; the others are aggregated variables obtained by factor analysis.
* Correlation is significant at the 0.05 level (one-tailed).

Table 9
Predictive models for the two error variables

Error	Regression equation	R^2	P
ERE	0.129 + 0.0014 ∗ Average task duration	0.20	0.003
DRE	0.844 − 0.021 ∗ Estimated total duration −0.044 ∗ Estimation goals definition −0.054 ∗ Systematic risk assessment −0.033 ∗ Reporting frequency	0.51	0.000

5.3. Predictive models

In order to understand the combined effect of those independent variables that were significantly correlated with the dependent variables, we carried out a stepwise linear regression analysis for the two estimation error measures. The resulting regression models appear in Table 9. The stepwise criteria were a level of significance of .05 to add and .10 to remove independent variables. All the coefficients in the two models were significant at the .05 level.

The relative effort estimation error (ERE) is explained by the average task duration. It seems that effort estimation is usually based on the tasks duration and the entire project duration and therefore these variables are the only ones that have significant explanatory power.

As we have seen earlier in the correlation tables, the relative duration estimation errors are mainly correlated with the estimation process management factors. The regression model supports these findings. The relative duration estimation errors (DRE) can be explained by the estimated total duration and the process related variables, estimation goals definition, systematic risk assessment and reporting frequency. The signs of the regression coefficients conform to the direction proposed by the research hypotheses. The higher is the estimator's proficiency, and the better the estimation process is conducted, the smaller are the estimation errors.

6. Testing of hypotheses

The hypotheses were tested by examining the correlation coefficients between the various variables of the framework, and the two measures of estimation errors. We also took into consideration the *T*-tests and the regression results.

H1. Projects with higher level of uncertainty exhibit larger duration and effort estimation errors

The results suggest partial confirmation of H1. It seems that more complex projects, with higher uncertainty, as indicated by the project size (duration, effort, and number of potential users), the complexity of the application and the level of innovativeness of need, are more likely to entail larger duration and effort errors. The negative correlations of estimated duration and implementation complexity with the relative duration error (−.518 and −0.290, respectively) indicate that for more complex (or larger) projects the duration estimation errors are smaller than the overall duration. The team level of experience has also an effect on

DRE; low level of experience significantly increases the duration estimation error. It is interesting to note that unclear project definition and instability are not correlated with any of the types of estimation errors.

H2. Projects with higher use of estimation development processes exhibit smaller duration and effort estimation errors

The correlation and regression analyses suggest partial confirmation of H2. While the number of WBS levels is not correlated with estimation errors, projects with shorter activity durations and smaller task efforts have smaller effort estimation errors. These findings support previous findings that decomposition of the project into smaller activities helps coping with effort estimation. Another interesting finding is that when estimation goals are better defined the estimation errors are smaller; the correlation with duration errors is negative and significant while the correlation with effort errors is negative but not significant. The other two factors are also negatively correlated with duration estimation, but not to a significant level.

H3. Projects with higher use of estimation management processes exhibit smaller duration and effort estimation errors

Here again, the correlation and regression analyses suggest partial confirmation of H3. Higher level of customer control helps to reduce the relative effort estimation errors, while more frequent reporting, team performance assessment and systematic risk management contribute to reducing duration errors. Looking at the factual variables shows that frequent work plan updates also contribute to better duration estimation. Frequent reporting creates higher commitment and, therefore, better estimations. Customer control via a steering committee also helps to reduce the relative effort estimation errors. Most of the significant correlations are associated with duration estimation, probably due to the greater emphasis on duration management than on effort management.

H4. Projects where the estimates were made by more experienced estimators will have smaller duration and effort estimation errors

The variable number of projects previously managed by the estimator is significantly correlated with DRE. The estimator's experience in the application area is also highly, but not significantly, correlated with DRE. Furthermore, the *T*-tests that were performed found that there is a significant difference between average estimation errors provided by estimators with training in IT project management and those without such training. When the estimators have some training in estimation techniques their average errors are smaller than those of untrained estimators. It seems that training in estimation techniques contributes mainly to improving duration estimations. Taking into account the correlation and *T*-test results, we can conclude that hypothesis H4 was also partially supported.

7. Discussion and recommendations

The results of our analysis suggest that duration and effort estimation errors increase as the project uncertainty increases, but at a declining rate. This is apparent from the negative correlation with the relative errors. The smaller relative errors in larger projects are probably a result of the higher management attention to large projects and the consequently intensive estimation process, which lead to better estimation accuracy.

The findings of this study suggest that project uncertainty has a stronger effect on duration estimation than on effort estimation errors. One possible explanation is that the organization studied is more focused on deadlines than on budgets, and consequently will take steps to reduce deviations in project duration, perhaps by adding more resources beyond the original effort estimates. Such steps will be more common in high uncertainty projects, ensuring adherence to the original completion date and reducing the uncertainty effect, on one hand, while increasing the effort estimation errors, on the other hand.

Our results support the claim in the literature that planning at a more detailed level (shorter activities, smaller tasks) contributes to the accuracy of the estimates and reduces the size of the estimation errors. In addition, using the estimates for progress control and for project team evaluation purposes increases the commitment of the estimators and reduces the occurrence of estimation errors. Additional project management practices that were found to be related to lower duration estimation errors include more frequent progress control, more frequent updating of work plans and assessing the risks in the project.

It is interesting to note that although a higher level of estimation management was associated with smaller duration estimation errors, there was no statistically significant relationship with effort estimation errors. This again suggests that estimation management practices in this organization are focused on duration rather than effort, possibly because meeting deadlines is perceived as being more important than meeting budgets or actual effort data are less available or reliable than duration data and the accuracy of the effort estimations is consequently reduced. This finding is consistent with the literature regarding the importance of managing the estimates throughout project execution.

We found in general that estimator experience contributes to reducing estimation errors. This relationship applies to experience in the specific application area of the project being estimated, while the total number of years of professional experience in IT projects turns out to be not relevant. Apparently the variability across applications is substantial, and experience acquired in one application area is not carried over to others. Furthermore, the variable that is highly correlated with estimation accuracy is the number of projects that the estimator has managed rather than the number of years of experience in managing projects. This is a reasonable finding since each project managed created an

additional opportunity to develop estimates during the planning phase, thus improving the estimator's skill and accuracy. A project manager becomes better at estimating tasks by managing many small projects than by managing fewer larger projects over a comparable time span. Overall, the effect of estimator experience is more significant for duration estimation than for effort estimation. This may yet again be a result of the priorities at the organization or of the difficulties inherent in developing and managing effort estimates as opposed to duration estimates.

8. Concluding remarks

The conclusions that can be drawn from this study are limited by the characteristics of the research methodology: the sample comes from a single, albeit typical, organization; the number of projects in the sample is relatively small, precluding a more detailed research model and more sophisticated statistical analyses; and there is always the issue of the validity of data measured on Likert scales. Nevertheless, the results obtained add some insight and better understanding of time and effort estimation processes and identify the main factors influencing estimation accuracy.

The findings suggest that effort estimation and duration estimation are driven by somewhat different processes, and that the respective errors are affected by different factors, which may reflect the priorities of the project stakeholders. Thus, future research on estimation errors should take into account the goals, strategies and constraints of the executing organization and possibly of the customer organization. Our findings support the intuitive notion, already documented in the literature, that estimation errors can be reduced by investing in more detailed planning. They also reinforce Hughes's [18] claim regarding the importance of managing the estimation process during project execution. Specifically, increasing the extent of control, team motivation and accountability appears to reduce estimation errors. Finally, our study was able to pinpoint the number of projects previously managed, rather than the number of years of experience, as the main driver of the estimator's experience that affects estimation accuracy.

From a practical perspective, the specific findings of this study can be used as guidelines for better duration and effort estimation on issues such as accounting for project uncertainty while managing expectations regarding estimate accuracy; investing more in detailed planning in order to reduce estimation errors; and selecting estimators based on the number of projects they have managed. Although the regression equations obtained here are valid only for the organization in the study, the same methodology can be adopted by any organization in order to benchmark their estimation performance.

We conclude with some thoughts on extending this line of research. Obviously, a larger sample including multiple organizations from different business areas and different cultures would be very helpful for validating and extending the findings of this study. In addition, it would enable the analysis of the differences between underestimations and overestimations, which was not undertaken here due to sample size limitations, as well as the effect of interactions among the various factors. A more refined research model that would include additional constructs, such as organizational culture and policies, stakeholder expectations and other factors that have been seldom studied in other contexts, might generate some new insights. Finally, it should be interesting to study the evolution of estimation errors over time in order to understand learning effects, both at the individual and organizational levels.

Appendix A. Questionnaire Items

1.1 *The project*

1.1.1 Project objectives were clear when the work plan was drawn up.

1.1.2 The means for achieving the objectives were clear when the project was planned.

1.1.3 The project was defined by management as a high priority project.

1.1.4 Project priority was downgraded during the course of the project.

1.1.5 The project was defined by management as a high urgency project.

1.1.6 The project significantly depends on outside software and services suppliers.

1.2 *The end-product (SW system)*

1.2.1 The system is based on new hardware not well known to the IS division.

1.2.2 The system is based on new software not well known to the IS division.

1.2.3 The system is based on more technologies than previous projects.

1.2.4 The system requires tight integration with other systems.

1.2.5 The system has to handle large quantities of data.

1.2.6 The system has to handle a massive amount of communications traffic.

1.2.7 The system has to carry out complex processing operations.

1.2.8 The system has to support high security levels.

1.2.9 The system is required to support complex work processes.

1.2.10 The system has to be robust with high level of availability.

1.3 *The project team*

1.3.1 The project team lacks experience or expertise in the development methodology followed by the project.

1.3.2 The project team lacks experience or expertise in the development environment.

1.3.3 The project team lacks experience or expertise in the specific application.

1.3.4 The project team is not sufficiently familiar with the user work processes.

1.3.5 The role of all project team members has been well defined.
1.3.6 Interpersonal relations among the team members are good.
1.3.7 The team is highly motivated.
1.3.8 All team members were assigned to the project at least 75% of their time.
1.3.9 The planned human resources were available when required.

1.4 *The users*
1.4.1 The project has various users with different needs.
1.4.2 The users are keen to start using the system.
1.4.3 Users' requirements were defined precisely and in great detail.
1.4.4 The users' requirements have changed substantially throughout the project.
1.4.5 The users provided the project with information on the required work processes.
1.4.6 The operational experts and the steering committee members are not sufficiently familiar with IS development processes.

2.1 *Developing the work plan*
2.1.1 The estimation served to establish the project's duration.
2.1.2 The estimation served to establish the required manpower resources.
2.1.3 The estimation served to plan project staffing.
2.1.4 The estimation served as the baseline for monitoring the project's progress.
2.1.5 The estimation served as the basis for evaluating development staff performance.
2.1.6 The estimation served as the basis for evaluating the estimator's performance.
2.1.7 A standard project organizational structure was used for project planning.
2.1.8 The estimation was carried out using only professional judgment, without the help of any additional tools or techniques.
2.1.9 Rules of thumb were used for duration and effort level estimates.
2.1.10 Data from previous projects in which you participated was used for estimation.
2.1.11 Data from previous projects done by the IS division was used for estimation.
2.1.12 The estimates were based on external standards (analysts, other organizations).
2.1.13 The estimation was carried out top-down.
2.1.1 The estimation was carried out bottom-up.
2.1.15 Team members' capabilities were known at the time of the estimation.
2.1.16 The project team members were involved in the estimation process.
2.1.17 Technical and organizational experts were used to carry out the estimation.
2.1.18 The development team was highly committed to the planning process.
2.1.19 Estimates were approved by the IS management.
2.1.20 Estimates were approved by the steering committee or the user's representative.
2.1.21 Estimates were adjusted to account for time and resources constraints dictated by management or by the user's representative.

2.2 *Managing relations with the parties involved*
2.2.1 IS management are informed of the project's progress and help solving problems.
2.2.2 The project has an effective steering committee.
2.2.3 Operational experts representing the user are effectively involved in the project.
2.2.4 There is an effective technical committee that supervises and coordinates all activities of the IS division.
2.2.5 The project enjoys the cooperation of technical support entities.
2.2.6 The project enjoys the cooperation of organizational support entities.
2.2.7 The project enjoys the cooperation of other projects that interface with the system.
2.2.8 The project enjoys the cooperation of external suppliers.

2.3 *Management tools and procedures*
2.3.1 Computerized tools are used for managing and supervising tasks beyond the level of the annual work plan.
2.3.2 A periodic status report is distributed to all stakeholders.
2.3.3 Updated status and estimation data are presented to the IS division technical committee.
2.3.4 Updated status and estimation data are presented to the customer's steering committee.
2.3.5 The work plan is updated when there is a substantial change in the project scope.
2.3.6 The work plan is updated at the end of every stage of the life cycle.
2.3.7 The project manager is evaluated by the degree to which the planned schedule and budget are met.
2.3.8 The members of the project team are evaluated by the degree to which the planned schedule and budget are met.

2.4 *Risk management – problems and lessons learned*
2.4.1 There is a defined and effective process for quickly addressing problems that arise during the course of the project.
2.4.2 There is an ongoing systematic process for managing risks.
2.4.3 There is an ongoing process for learning lessons and implementing the appropriate changes.

3.1 *Estimator experience*
3.1.1 Did you have experience with the development methodologies used in the project?

3.1.2 Did you have experience with the development environment used in the project?
3.1.3 Did you have experience with projects of similar types?
3.1.4 Were you familiar with the operational processes of the project's users?

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